

	Work Instruction	Doc. Revision	03
	AXI V810 XXL Filter Height Sensor Adjustment Procedure	Effective Date	10 Jan 2017

This procedure is applicable to below models:

V810 STD	☐	S1	☐	S2
V810 XXL	☒	S1	☒	S2
V810 S2EX	☐			
V810 Mini	☐			

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	1 Dec 2016	First Release	CHIN CHEE YAN
01	30 July 2016	Second Release	MUHAMMED SHAMEEM
02	20 Dec 2016	Third Release	MUHAMMED SHAMEEM
03	10 Jan 2017	Template Changes & Content Update	ONG KEAN SHENG

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A. Filter Height Sensor Adjustment

Remarks : Check the Filter Height Sensor Amplifier Setting at Section B before proceed with Filter Height Sensor Adjustment

1. Prepare Variable Magnification Filter Jig Assembly (Part ID: 4LASS-A265) as Figure 1



Figure 1 4LASS-A265 XXL Filter Height Jig

2. Go to Services->Subsystem Status & Control-> Panel Handling and click Initialize as Figure 2.

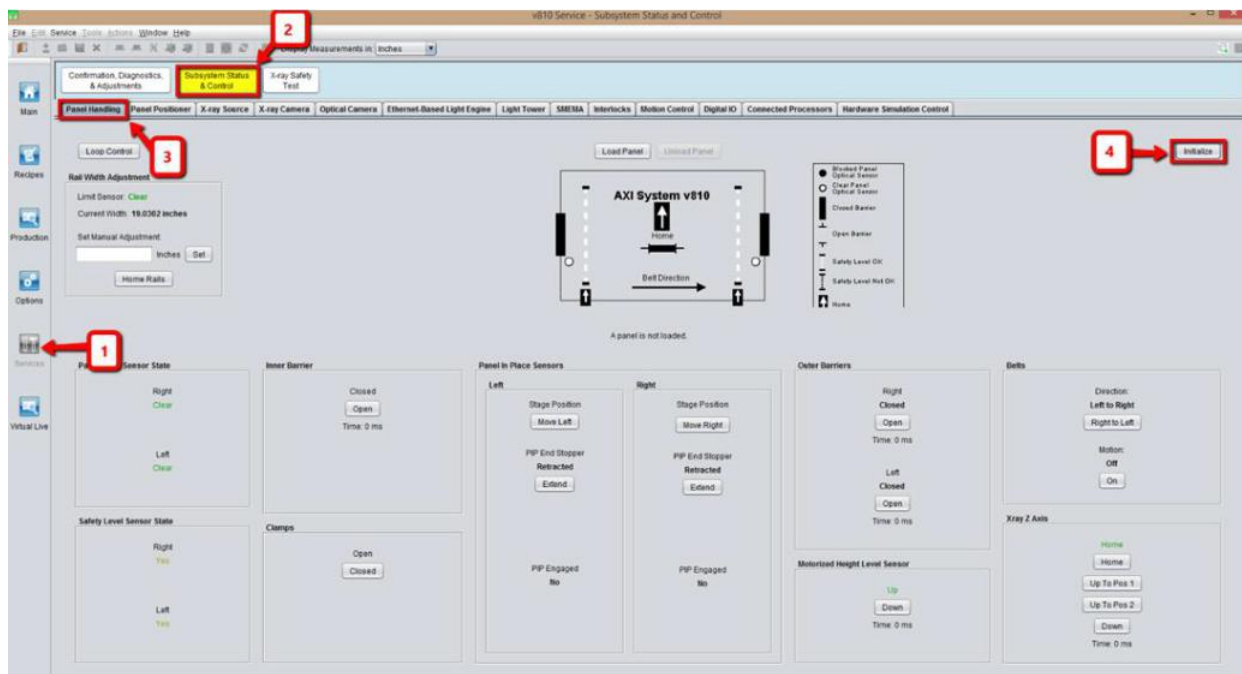


Figure 2 Panel Handling

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- Click "Load Panel", Enter the width & length as 26 inch as Figure 3.
Click "Load" as Figure 3.

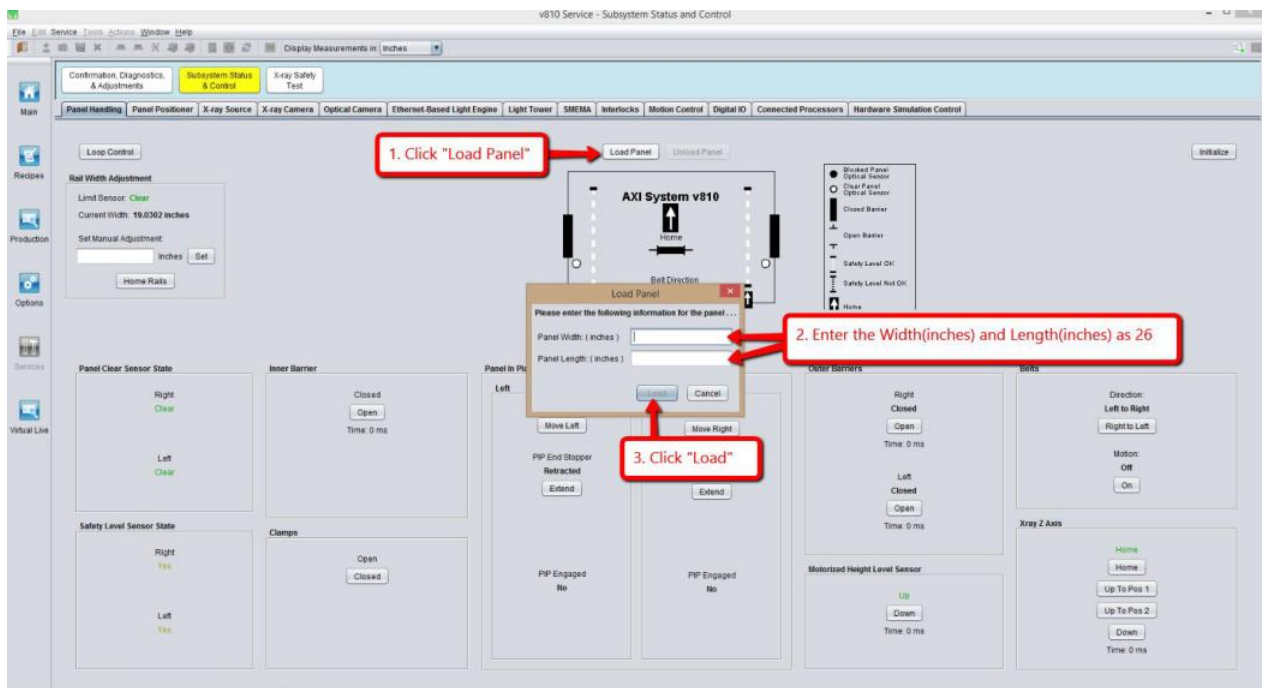


Figure 3 Panel Handling (Loading method)

- Once outer barrier open, click "Cancel". The outer barrier will close immediately.
Manually open the left outer barrier using the Panel Handling as Figure 4.

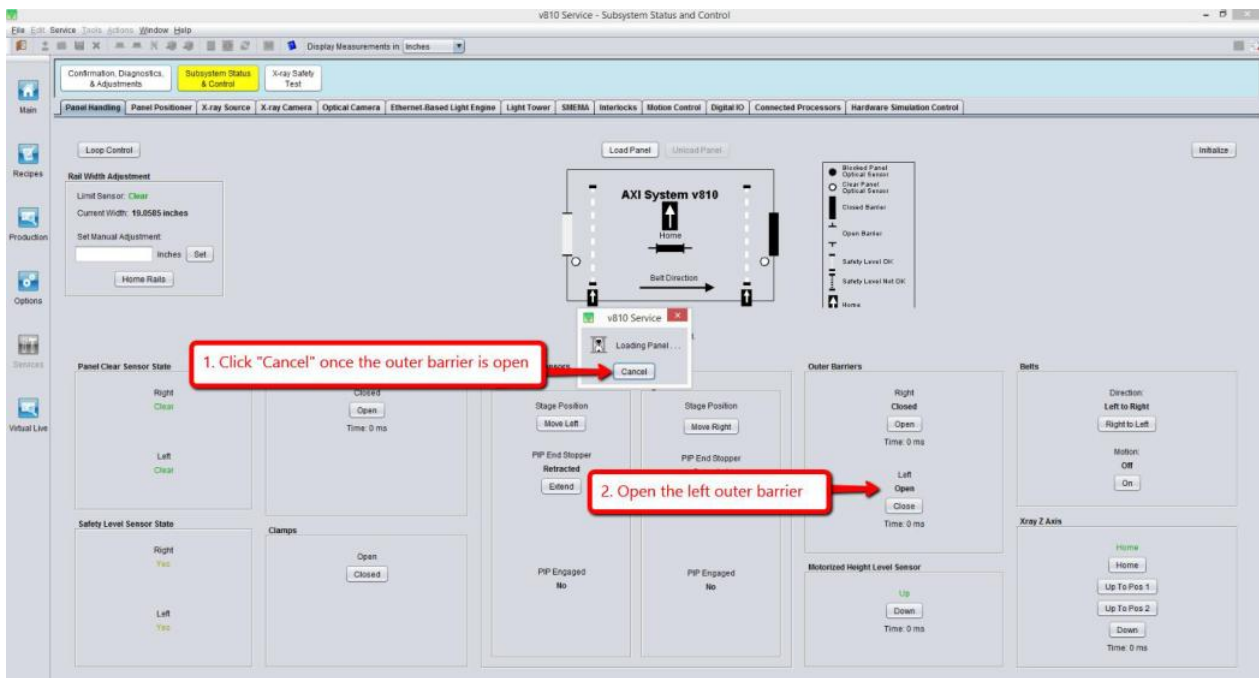


Figure 4 Panel Handling (Open Outer Barrier)

- Close X-ray source and open front access door.

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6. Place 4LASS-A265 near to the edge of PIP Stopper as Figure 5.



Figure 5 4LASS-A265 (Filter Height Jig Location)

7. Once Filter Height Jig is align as Figure 6, close Clamper by using the V810 GUI to hold the Jig in its position

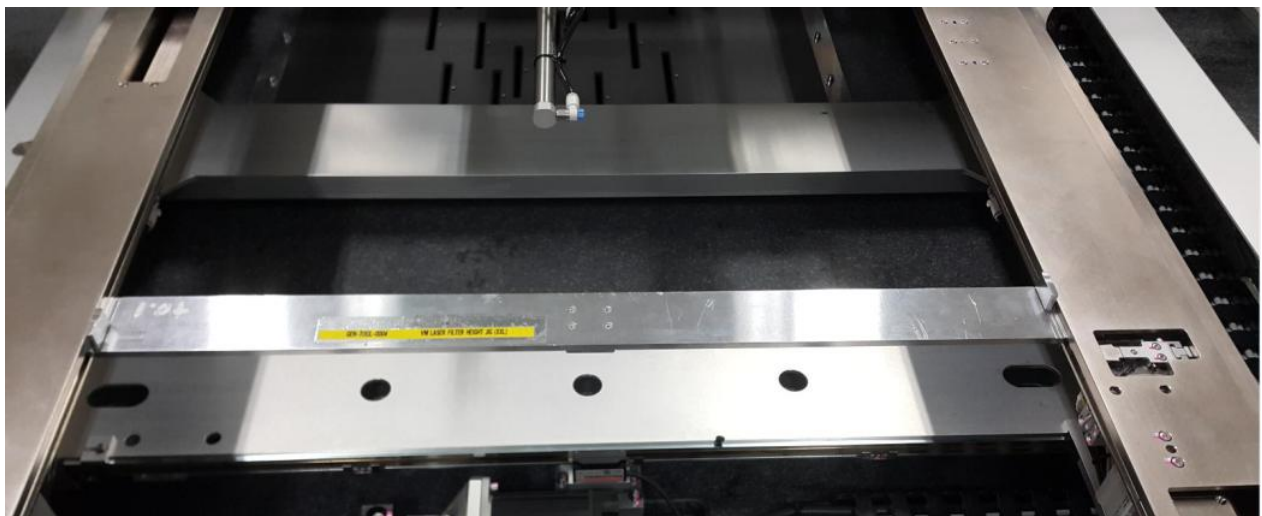


Figure 6 Filter Height Jig Placement (Clamper)

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8. Remove SHCS M3x10 (4x) to access the filter height sensor head as Figure 7.

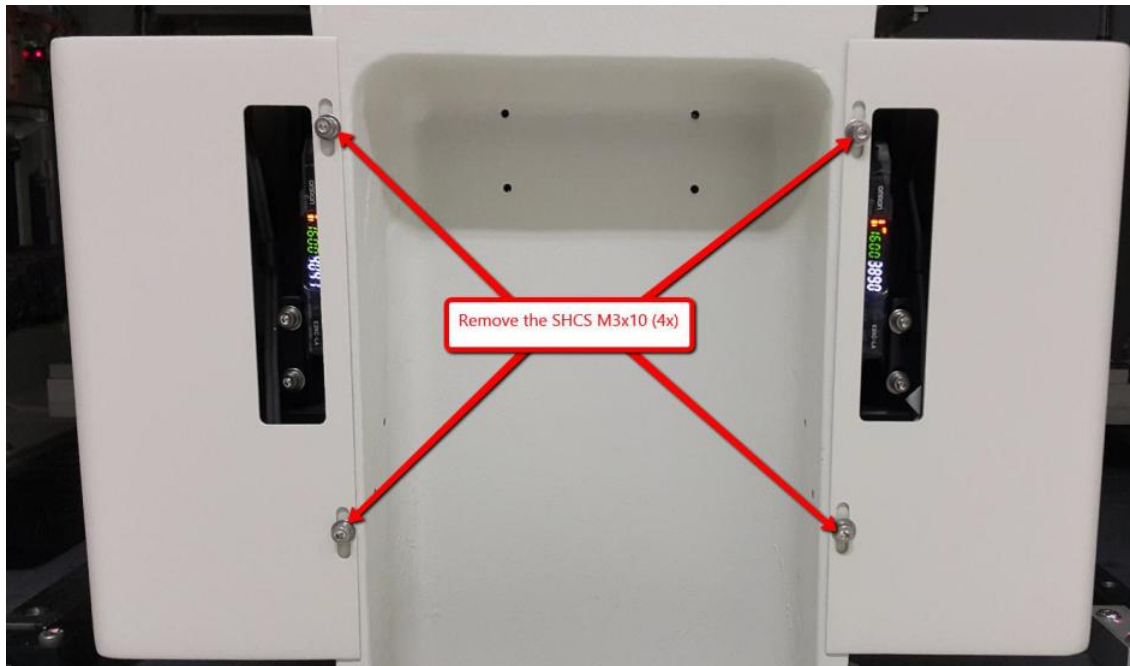


Figure 7 Filter Height Sensor Cover Bracket

9. Loosen 2 x screws at Sensor Head as Figure 8. Adjust Sensor Head until laser spot is pointing at middle of the reflector as Figure 9.

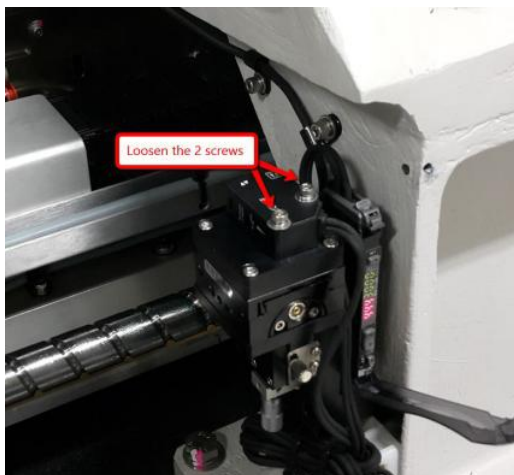


Figure 8 Sensor Head

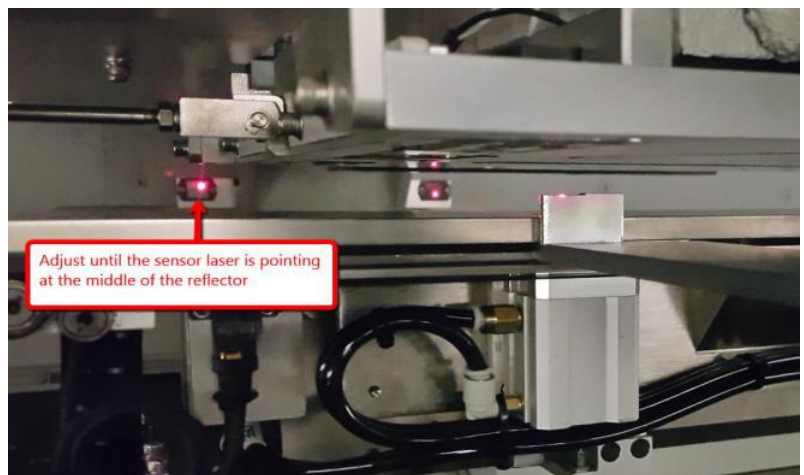


Figure 9 Laser Pointing Direction

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10. Manually move the stage by turning the X-Axis Coupling until Filter Height Laser point at Variable Mag Jig as Figure 10.

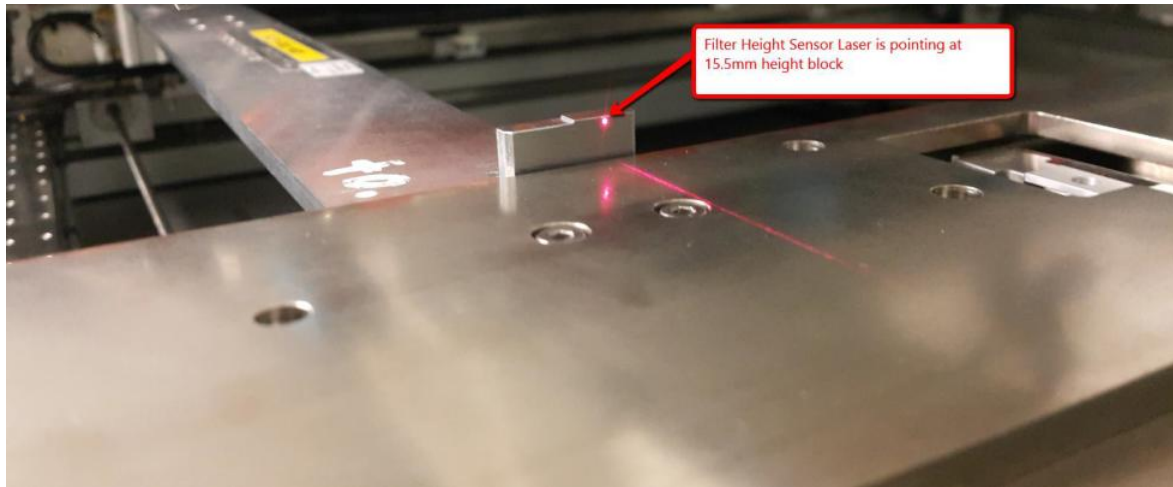


Figure 10 Filter Height Jig

11. Loosen the screw using the Allen Key as Figure 11. Secondly, loosen the Lock Screw as Figure 11. Adjust the height of the filter height sensor by turning the micrometer knob in the Figure 11 until the laser spot is pointing at the middle of the 15.5mm block (refer to the Figure 12)



Figure 11 Filter Height Sensor Module

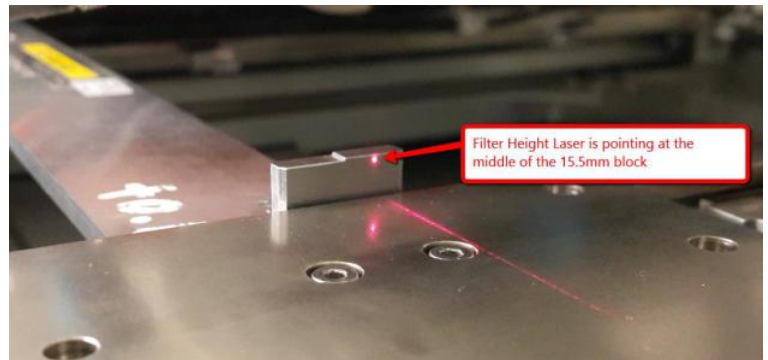


Figure 12 Filter Height Sensor Laser at the 15.5mm

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12. Perform below adjustment simultaneously until threshold reading between Front Block(15.5mm) & Rear block (15.5mm) are within the desire variance (Variance Spec: 50)

Eg :Front Block Threshold-1600 ; Rear Block Threshold - 1592, Variance : 8

- Adjust Sensor Block Height - Step 11
- Adjust Sensor Angle as Figure 14.

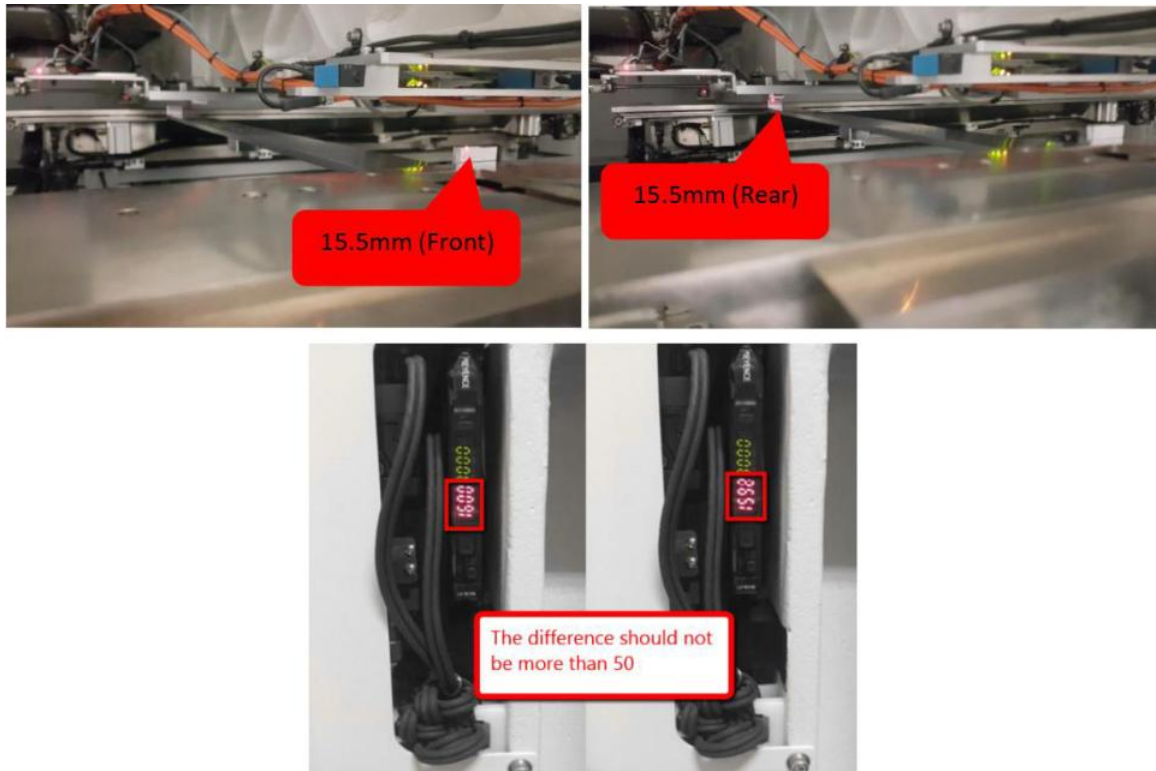


Figure 13 Sensor Laser Adjustment

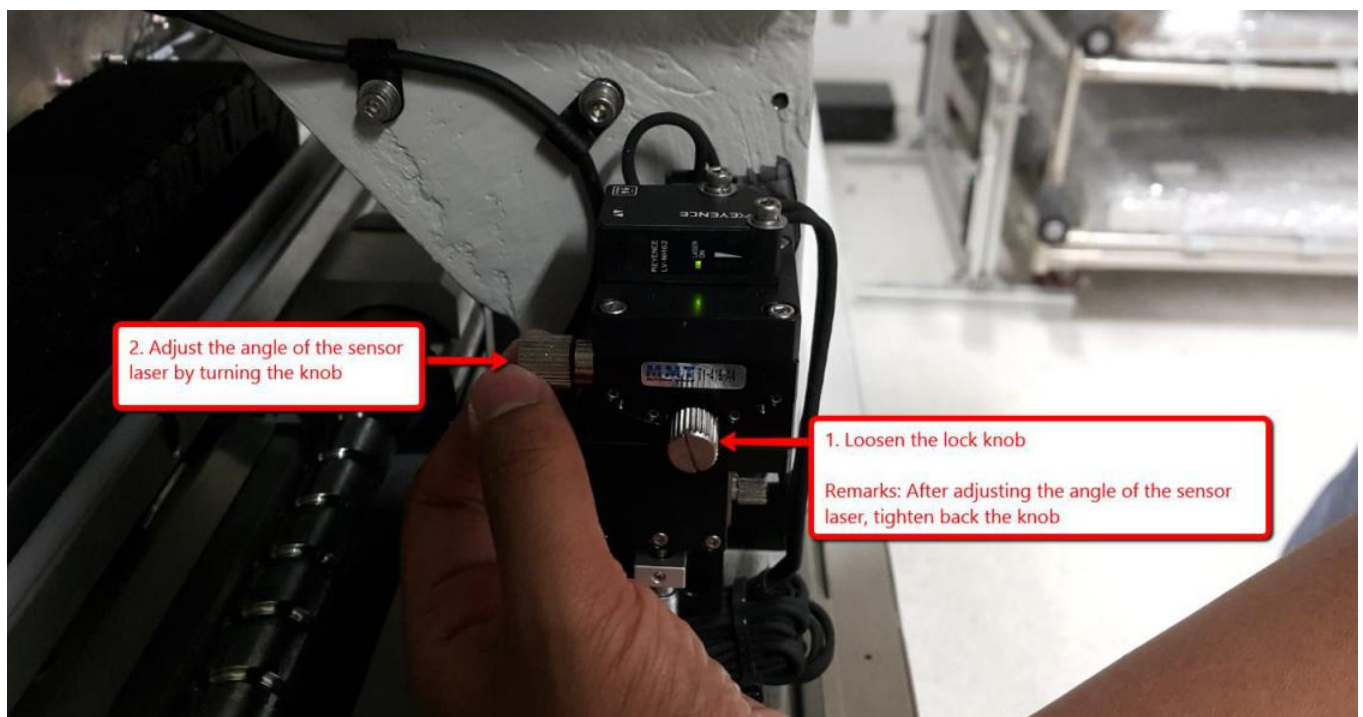


Figure 14 Sensor Laser Angle Adjustment

13. Repeat Step 10~13 until desire result is within below spec (Table 1).

Orientation	Jig Height	Amplifier Value
A (Fix Rail)	15.0	>600
	15.5	<300
B (Adj Rail)	15.0	>600
	15.5	<300

Table 1 Filter Height Sensor Adjustment Values

14. May refer to the below diagram for filter sensor spot beam angle leveling elaboration.



Figure 15 Filter Sensor Spot

- ✓ If Fix Rail's block intensity is lower than Adjustable Rail's block intensity, eventually indicate current laser spot beam is pointing upwards. Require to adjust the Filter Height Sensor to upper position and lower down the beam angle.
- ✓ If the Fix Rail's block intensity is higher than Adjustable Rail's block intensity, eventually indicate current laser spot beam is pointing downwards. Require to adjust the Filter Height Sensor to lower position and adjust sensor beam into upward angle.

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15. Verify the sensor status by refer to below table.

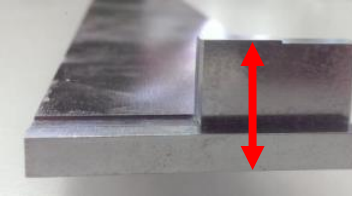
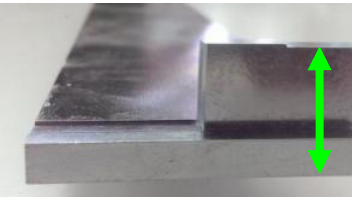
Jig Height Image	Height	Filter Sensor Status	DIO Input Bit	Term
	15.5			➤ Only remain 1 LED at sensor is emitted. ➤ DIO input bit show “Block”. ➤ Amplifier Value (Red) less than 300
	15			➤ 4 LED at sensor are emitted. ➤ DIO input bit shown “Clear”. ➤ Amplifier Value (Red) more than 600

Table 2 XXL Filter Height Sensor Adjustment

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17. After Left Filter Height Sensor Adjustment complete, unclamp the clumper at Panel Handling tab.
18. Re-position the 4LASS-A265 near to the edge of the Right PIP Stopper as Figure 16.



Figure 16 Right PIP Stopper

19. Align the jig as Figure 17, close the clumper at Panel Handling Tab to hold the jig in its position.



Figure 17 Right PIP Sensor Stopper

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20. Manually move the stage by turning the X-axis coupling until Filter Height Sensor Laser point at the Variable Mag Jig as shown in the Figure 18.

Repeat Step 10 ~Step 16 for Right Filter Height Sensor Module.



Figure 18 Right Filter Height Sensor

21. After adjustment complete, un-clamp the clumper and remove the Jig from the machine
22. Assemble Filter Height Sensor Cover to its respective module and store the filter height jig at respective place.

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B. Filter Height Sensor Amplifier Configuration

- Figure 1 shows the each button and the function of the sensor amplifier

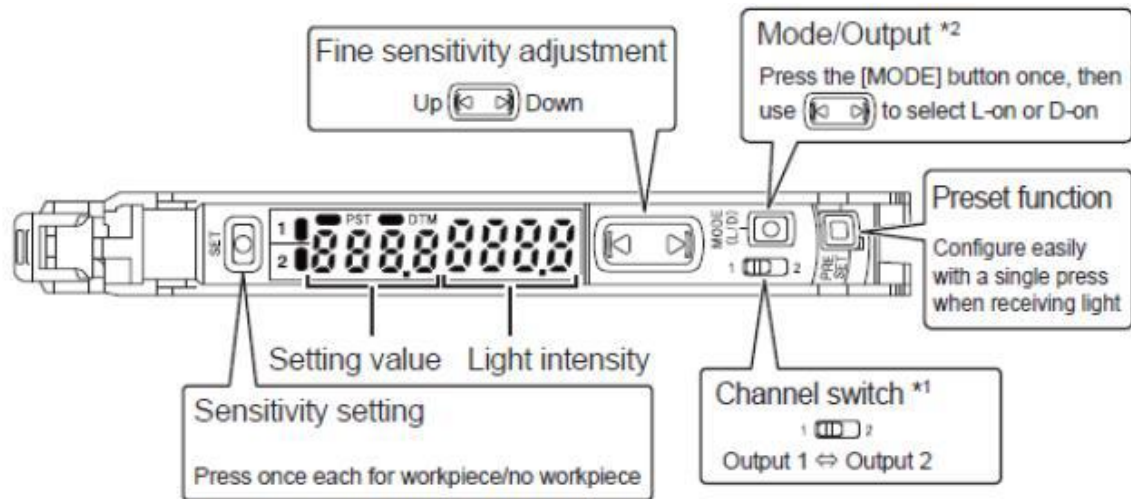


Figure 19 Keyence Sensor Amplifier

- Open amplifier top cover, ensure the selected channel is Channel 1
- Press the “Mode” button and hold the button for more than 3 seconds to enter the setting mode as shown in the Figure 20

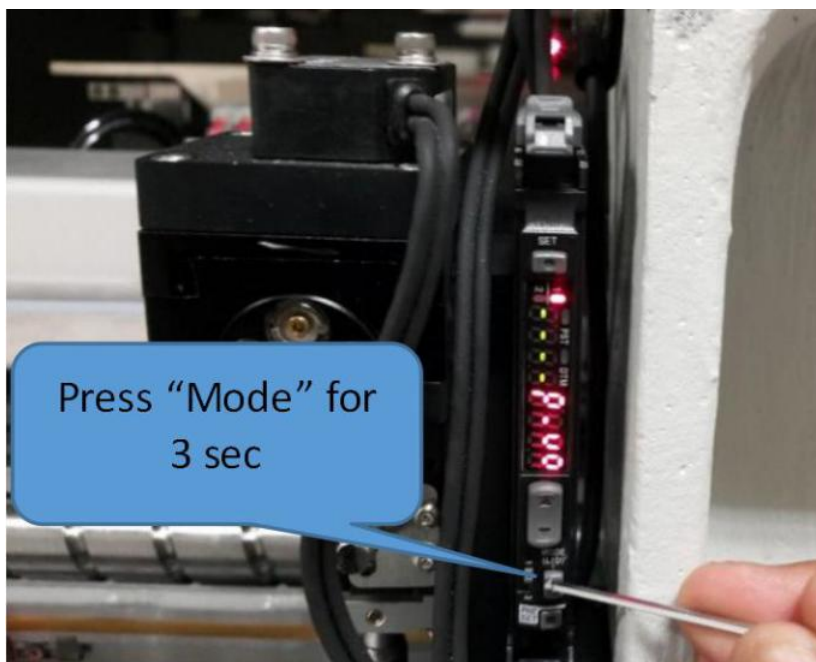


Figure 20 Keyence Amplifier Mode Setting

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4. Press “Up” or “Down” until the “SuPr” mode is selected as shown in the Figure 21.

Remarks : Maximum intensity can reach more than 1100 when the laser is directly shooting at the reflector. If the reading is less than 1100, clean the reflector surface as shown in Figure 22



Figure 21 SuPr Mode

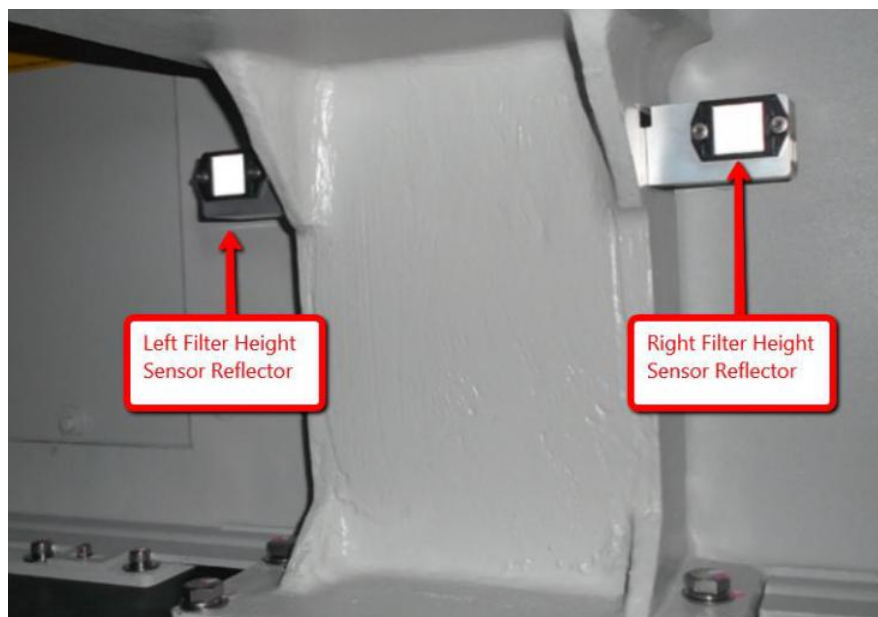


Figure 22 Filter Height Sensor Reflector

5. Press the “mode” button followed by “Up” or “Down” until the “Std” option is selected as shown in the Figure 23



Figure 23 "Std" Mode

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6. Press the “mode” button followed by “Up” or “Down” button to select the “Func” as shown in the Figure 24



Figure 24 "Func" Mode

7. Press the “mode” button followed by “Up” or “Down” and select the “on-d” option as shown in the Figure 25



Figure 25 "on-d" mode

8. Press the “mode” button followed by “Up” or “Down” button until value reach 5000 as shown in the Figure 26



Figure 26 Delay timer

9. Press the “mode” button followed by “Up” or “Down” button to select the “dEtc” mode as shown in the Figure 27. Press “mode” button again followed by “Up” or “Down” button to select the “Std” option as shown in the Figure 27.



Figure 27 Std mode

10. Press the “mode” button followed by “Up” or “Down” to select the “In off” mode as shown in the Figure 27.



Figure 28 In off mode

11. Press “mode” button again to exit the setting
12. Press “L/D” button followed by “Up” or “Down” button to select the “L-on” mode as shown in the Figure 29

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Figure 29 L-on Setting

13. After few seconds, the amplifier will return to the main menu

Set the threshold for the sensor amplifier to 450 using “Up” or “Down” button as shown in the Figure 30



Figure 30 Threshold setting